



Sentiment Analysis of Hotel Reviews Using Machine Learning Techniques

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Abstract. Sentiment analysis is the task of identifying opinions expressed in any form of text. With the widespread usage of social media in our daily lives, social media websites became a vital and major source of data about user reviews in various fields. The domain of tourism extended activity online in the most recent decade. In this paper, an approach is introduced that automatically perform sentiment detection using Fuzzy C-means clustering algorithm, and classify hotel reviews provided by customers from one of the leading travel sites. Hotel reviews have been analyzed using various techniques like Naïve Bayes, K-Nearest Neighbor, Support Vector Machine, Logistic Regression, and Random Forest. An ensemble learning model was also proposed that combines the five classifiers, and results were compared.

Keywords: Sentiment analysis · Sentiment detection · Sentiment classification · Machine learning · Tourism

1 Introduction

Nowadays, people generally prefer to communicate and socialize on the web. On travel sites, users express their opinions and write reviews of their hotel experience. Taking advantage of these huge volumes of data is of great value to tourism associations and organizations which aim to increase profitability and enhance or maintain customer satisfaction. Customer Reviews provided through the text are considered either subjective or objective. Sentences with subjective expressions include opinions, beliefs, personal feelings and views, while objective expressions include facts, evidences and measurable observations [1]. Most of the sentiment analysis approaches apply sentiment detection first which differentiates between objective and subjective reviews then determines the sentiment polarity of subjective reviews whether it is positive or negative. Fuzzy C-Means clustering algorithm was applied for sentiment detection to classify sentences to subjective or objective. Objective sentences are filtered out and only subjective meaningful sentences are retained.

We used five different machine learning classifiers for sentiment classification namely Naïve Bayes, K-Nearest Neighbor, Support Vector Machine, Logistic Regression and Random Forest. Then we applied an ensemble learning model between the five classifiers to achieve better results. This paper is organized as follows. First, a summary of some related work is found in Sect. 2. Section 3 introduces the proposed

system and discusses the techniques used throughout this paper. Section 4 shows the final results and compares the performance of various techniques used. Finally, Sect. 5 presents the conclusion.

2 Related Work

The main goal of sentiment analysis is to identify the polarity of text whether it is positive or negative. Recently, data available online related to tourism is increasing exponentially [2]. Sentiment analysis can effectively aid decision making in tourism, by improving the understanding of tourist experience [3].

Many machine learning methods are used for sentiment classification of hotel reviews. Machine learning methods are categorized into supervised, semi-supervised and unsupervised approaches. Neha Nandal et al. [4] have utilized the Support Vector Machine to classify Amazon customer reviews, where aspect terms are identified first for each review and finally give a polarity score for each review. Their outcomes showed that among the three kernels of the Support Vector machine, the Radial Basis Function (RBF) kernel provided the best result. Vishal S. et al. [5] have performed sentence-level sentiment analysis. They mainly focused on negation identification from online news articles. They used machine learning algorithms like Support Vector Machine and Naïve Bayes. They achieved accuracies of 96.46% and 94.16% for Support Vector Machine and Naïve Bayes respectively. Aurangzeb Khan et al. [6] proposed a sentence-level sentiment analysis approach. They extracted subjective sentences and labeled them positive or negative based on their word-level feature using the Naïve Bayes classifier. Then they applied the Support Vector machine for sentiment classification. Their proposed method on average achieves an accuracy of 83%. Vishal Kharde et al. [7] showed that machine learning methods, such as Support Vector Machine and Naïve Bayes have the highest accuracy in sentiment classification on twitter data. Xia et al. [8] used an ensemble framework for Sentiment Classification that combined various feature sets namely Part-of-speech information, Word-relations, and three machine learning techniques, which are Naïve Bayes, Maximum Entropy, and Support Vector Machine. They found that Naïve Bayes worked better on feature sets with a smaller size. By contrast, the Maximum Entropy and Support Vector machine were more effective for high-dimensional feature sets. They performed different ensemble approaches like the fixed combination, weighted combination, and Meta-classifier combination to improve the accuracy.

Rehab M. Duwairi et al. [9] performed sentiment analysis on Arabic reviews using machine learning methods. Three classifiers were used which are Naïve Bayes, Support Vector Machine and K-Nearest Neighbor. Their experimental results showed that the Support Vector machine achieved the highest precision, while the K-Nearest Neighbor achieved the highest recall. Anjuman Prabhat et al. [10] Performed sentiment classification on twitter reviews. They have used Naïve Bayes and Logistic Regression for the classification of reviews. The results showed that Logistic regression achieved better accuracy and precision than the Naïve Bayes. Bhavitha et al. [11] compared the performance of the Random Forest and Support Vector Machine on sentiment analysis. Random Forest obtained better accuracy, but requires higher processing and training

time. Gayatri Khanvilkar et al. [12] employed both Random Forest and Support Vector Machine on sentiment analysis for sentiment classification. They claim that their proposed system helps to improve Sentiment analysis for Product Recommendation, using Multi-class classification.

One of the main tasks in sentiment analysis is sentiment detection. Sentiment detection or subjectivity detection is the task of identifying subjective text. In this work, Samir Rustamov et al. [13] used the Fuzzy Control system and Adaptive Neuro-Fuzzy Inference System for sentence-level subjectivity detection from movie reviews. They used informative features that improve the accuracy of the systems with no language-specific requirements. Iti Chaturvedi et al. [14] used both Bayesian networks and fuzzy recurrent neural networks for subjectivity detection. Bayesian networks are used to capture dependencies in high-dimensional data, and fuzzy recurrent are then used to model temporal features. They claim that their proposed model can deal with standard subjectivity detection problems, and also proved its portability across languages.

3 Proposed System

Initially, the dataset used contains 38,932 labeled hotel reviews from the Kaggle website. This data set provides reviews of a single hotel. Some pre-processing was done to clean and prepare data for sentiment analysis (Fig. 1). Pre-processing of text involves eliminating irrelevant content from text [15]. Online texts usually contain lots of noise, and uninformative parts like typos, bad grammar, URLs, Stop words, and Expressions. The main goal of data pre-processing is to reduce the noise in the text, which should help improve the performance of the classifier, and speed up the classification process.

One of the steps of pre-processing is the removal of stop words. Stop words such as “the”, “a”, “an”, and “in” take valuable processing that is not needed so they should be removed. Removal of punctuations is another important step in the process of data cleaning. For example: “.”, “,”, “?” are important punctuations that should be retained while others need to be removed. Moreover, to avoid word sense disambiguation, an apostrophe lookup is required to convert any apostrophe into standard text. For example, “it’s a very nice place” should be transformed into “it is a very nice place”. Sometimes words are not in proper formats therefore standardizing words is important for data cleaning, for example: “it is a good place” should be “it is a good place”.

For feature extraction word embedding was used; to represent words in each sentence of a review. Word embedding is a technique where words or expressions are mapped to vectors of real numbers [16]. Word2vec technique [17] was employed for feature extraction. The corpus used in training the word2vec model was built with all the words available in the dataset. Word2Vec is one of the most popular methods that uses shallow neural networks. Word2vec can extract deep semantic features between words [17]. It computes continuous vector representations of words. The computed word vectors retain a huge amount of syntactic and semantic regularities present in the language [18], and transform them into relation offsets in the resulting vector space. The number of features extracted for each word is 100.

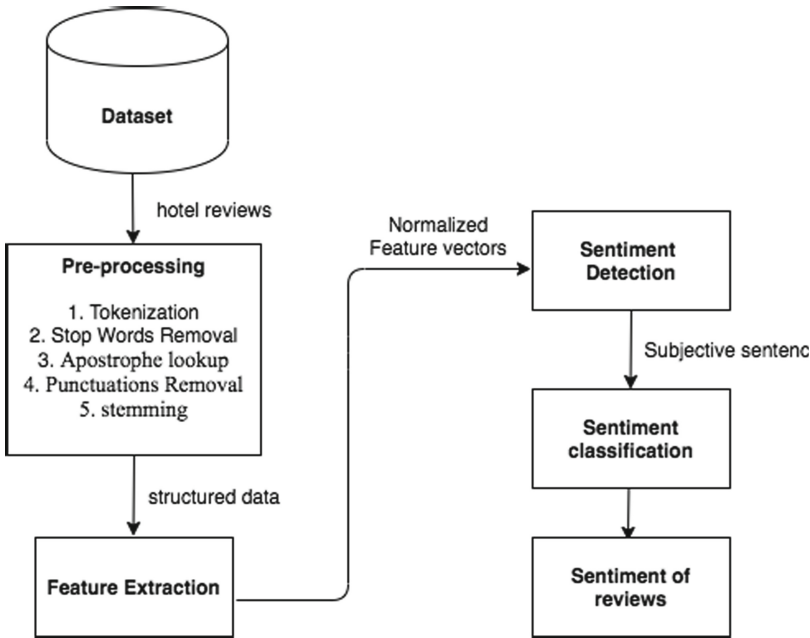


Fig. 1. The proposed system

Each review is composed of several sentences. After tokenizing sentences into words, the summation of all word vectors in the review is computed. Normalizing the features was a prior step to subjectivity detection. For subjectivity detection, the Fuzzy C-means clustering algorithm was used, to classify data into positive, negative, and objective. The Fuzzy C-means clustering algorithm is very similar to the K-means algorithm, but instead of using hard clustering where each data point can belong to only one cluster, Fuzzy C-means performs soft clustering in which each data point can belong to multiple clusters to a certain degree by means of a Membership Function. For example, data points that are close to the center of a cluster will have a high degree of membership for that cluster while data points that are far from the center of that cluster will have a low degree of membership. After clustering, objective sentences are discarded, and only sentences classified as positive or negative are retained for further classification. Five different machine learning techniques have been used to build our sentiment classification model, which are Naïve Bayes, K-Nearest Neighbor, Support Vector Machine, Logistic Regression, and Random Forest. An Ensemble learning model between the five classifiers was also applied to enhance the accuracy.

3.1 Machine Learning Methods

Naïve Bayes is the one of simplest models that is based on the Bayes Theorem. It works efficiently with large datasets as it is fast and accurate with a very low computational cost [19]. Naïve Bayes classifier assumes that all the features are

independent [20], but this assumption is not always true in real-life situations. However, Naïve Bayes often works well in practice. Given a predictor X , $P(C|X)$ is the posterior probability of class C and $P(X|C)$ is the probability of X given the class. $P(X)$ is the prior probability of X .

$$P(C|X) = P(X|C) * P(C)/P(X) \quad (1)$$

K-Nearest Neighbor (KNN) is also a very simple and easy-to-implement method [20]. The KNN algorithm assumes that similar features or data points exist close to each other. The most commonly used method for calculating the distance between two points is the standard Euclidean distance. However, there are other ways of calculating distance, and the choice depends on the problem. The input consists of the K closest training samples in the feature space. To choose the right value for K , we run the KNN algorithm several times with different values of K and choose the K that returns the highest accuracy. The Nearest Neighbors have been successful in many classification and regression problems but the main disadvantage of KNN that it becomes significantly slower when the volume of data increases.

Support Vector Machine (SVM) is one of the most effective and famous classification machine learning methods. SVM has been developed from statistical learning theory [19]. Moreover, it is memory efficient as it uses a subset of training points in the prediction process. SVM works well with a clear margin of separation and with high dimensional data. On the other side, it works poorly with overlapping classes and is also sensitive to the type of kernel used.

Logistic Regression is a statistical method that is used for binary classification. Logistic Regression measures the relationship between the test variable and our features, by estimating probabilities using its underlying Sigmoid function. The Sigmoid function is an S-shaped curve that can take any real-valued number and map it into a value between the range of 0 and 1, these values are then transformed into either 0 or 1 using a threshold classifier. It is well-known for its efficiency as it does not require too many computational resources, it is highly interpretable, it does not require input features to be scaled, it does not require any tuning, and it is easy to implement. A disadvantage of it is that it cannot solve non-linear problems since its decision surface is linear and it is also vulnerable to overfitting.

Random Forest consists of a large number of individual decision trees that operate as an ensemble. Each tree in the Random Forest returns a class prediction and Random Forest makes decision based on the majority of votes. The ensemble learning technique of this algorithm reduces the overfitting problem in decision trees and improves the overall accuracy [21]. It is very stable and can handle missing values and non-linear parameters efficiently. Random Forest is also comparatively less impacted by noise. The disadvantages are the long training time and high complexity as it generates a lot of trees that requires more computational power and resources than the simple decision tree.

4 Results and Analysis

Dataset of hotel reviews from the Kaggle website, contains 26,521 positive reviews and 12,411 negative reviews. After subjectivity detection, the total number of subjective reviews is 37,827 reviews. Sentiment classification performance was evaluated using precision, accuracy, recall, and F-score measures. Precision is the ratio of true positives to the total number of predicted positives, while Recall is the ratio of true positives to the total number of true positives and false negatives. True Positives (TP) are actually positive data points that are correctly classified as positive by the model, and False Negatives (FN) are positive data points that are misclassified as negative by the model. F-score is the weighted average of Precision and Recall, taking both false positives and false negatives into account. According to those measures, a comparison between the applied machine learning methods was applied, and results are presented in Table 1.

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) \quad (2)$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) \quad (3)$$

$$\text{F-score} = 2 * (\text{Recall} * \text{Precision}) / (\text{Recall} + \text{Precision}) \quad (4)$$

Table 1. Evaluation of methods performance.

Method	Accuracy	Precision	Recall	F-score
Naïve Bayes	0.718	0.727	0.957	0.655
K-Nearest Neighbor	0.838	0.836	0.956	0.827
Support Vector Machine	0.863	0.875	0.938	0.859
Logistic Regression	0.859	0.867	0.942	0.854
Random Forest	0.846	0.848	0.95	0.838

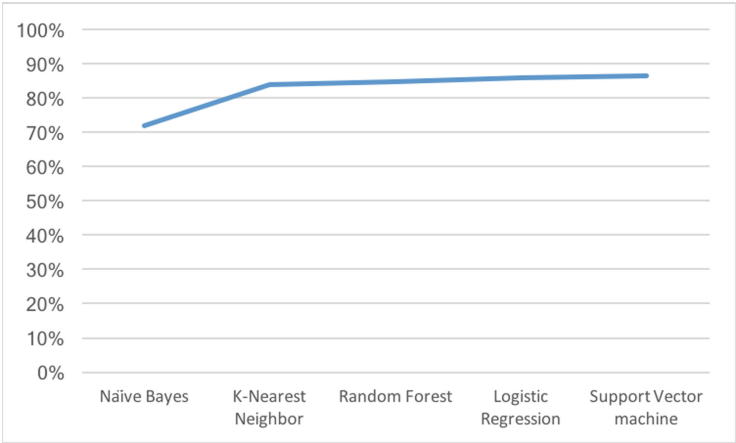


Fig. 2. Accuracy of various techniques

The results obtained show that Support Vector Machine and Logistic Regression techniques achieved better than the other techniques in terms of accuracy. For K-Nearest Neighbor, there is no ideal value for the initial number of neighbors K, it is selected after testing and evaluation. However, the best results were achieved when using 17 as K. Naïve Bayes classifier had the least accuracy compared to the other techniques (Fig. 2).

One of the problems that could cause errors in the classification process is that a single review can have different or mixed opinions on different aspects, which makes the task of assigning an overall polarity to the review hard. To achieve better results, an ensemble model of all classifiers was taken into consideration. An ensemble learning model is commonly known to outperform the performance of single classifiers. Ensemble models provide higher consistency and reduce errors. The results of the ensemble classifier are shown in Table 2. However, The Ensemble model didn't improve the accuracy in our case compared to the accuracy achieved by the Support Vector Machine.

Table 2. Evaluation of ensemble model.

Method	Accuracy	Precision	Recall	F-score
Ensemble learning model	0.862	0.872	0.94	0.858

5 Conclusion

In this research, different techniques were investigated to classify hotel reviews. The algorithms applied were Naïve Bayes, K-Nearest Neighbor, Support Vector Machine, Random Forest, and Logistic Regression. The best results were obtained by the Support Vector Machine. On the other hand, the Naïve Bayes classifier had the least achieved accuracy. The Support Vector Machine achieved 86.3% accuracy, logistic Regression achieved 85.9%, Random Forest classifier accuracy was 84.6% while K-Nearest Neighbor and Naïve Bayes classifier accuracies were 83.8% and 71.8% respectively.

An ensemble learning model was also created, combining those five classifiers to increase the accuracy. But after testing the model, the ensemble learning model achieved 86.2% accuracy, which means that the accuracy did not improve, and the Support Vector Machine is more efficient in this case.

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